



I. Inner product spaces, angle between vectors, Length and Orthogonality

1. Verify that $\mathbb{R}^n$ with $\langle \cdot, \cdot \rangle$ defined as the dot product is an inner product space. Also Expand $\langle 5u_1 + 8u_2, 6v_1 - 7u_2 \rangle$ and $\|2u - 3v\|^2$.
2. Let $V = \mathbb{R}^2$ be a vector space with $u = (u_1, u_2)$ and $v = (v_1, v_2)$. Consider the mapping $\langle \cdot, \cdot \rangle: V \times V \rightarrow \mathbb{R}$ defined by $\langle u, v \rangle = u_1v_1 - u_1v_2 - u_2v_1 + 4u_2v_2$. Is $(V, \langle \cdot, \cdot \rangle)$ an inner product space? Justify. If not, identify the properties that are satisfied and that are not.
3. Let $V = \mathbb{R}^2$ be a vector space with $u = (u_1, u_2)$ and $v = (v_1, v_2)$. Consider the mapping $\langle \cdot, \cdot \rangle: V \times V \rightarrow \mathbb{R}$ defined by $\langle u, v \rangle = u_1v_1 - u_1v_2 + u_2v_1 + 4u_2v_2$. Is $(V, \langle \cdot, \cdot \rangle)$ an inner product space? Justify. If not, identify the properties that are satisfied and that are not.
4. Let V be a vector space of all matrices of size $m \times n$, then verify whether the mapping $\langle \cdot, \cdot \rangle: V \times V \rightarrow \mathbb{R}$ given by $\langle A, B \rangle = \text{trace}(B^T A)$ is an inner product or not.
5. Check whether $\langle (x_1, x_2), (y_1, y_2) \rangle = x_1y_1 + 2x_2y_2 + x_1y_2 + x_2y_1$ defines an inner product on $\mathbb{R}^2$. If not, give a counter example vector where it fails.
6. Verify that the inner product on $V = \mathbb{C}^2$ defined by Hermitian inner product $\langle x, y \rangle = x_1 \text{conj}(y_1) + x_2 \text{conj}(y_2)$ for vectors $x = (x_1, x_2)$ and $y = (y_1, y_2)$ forms an inner product space. Also compute the inner product $\langle x, y \rangle$ for $x = (1 + i, 2)$ and $y = (2, i)$.
7. Let $V = \mathbb{R}^2$ be a vector space. For vectors $u, v \in \mathbb{R}^2$ define a function $\langle u, v \rangle = u^T A v$, where $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is a 2×2 real matrix. Determine the necessary and sufficient conditions on the entries a, b, c, d such that the function $\langle u, v \rangle$ defines a valid inner product on $\mathbb{R}^2$. Using these conditions obtained, determine whether the function defines an inner product when $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$.
8. If $u = (1, 3, -4, 2), v = (4, -2, 2, 1)$ and $w = (5, -1, -2, 6)$ in $\mathbb{R}^4$ then verify that $\langle 3u - 2v, w \rangle = 3\langle u, w \rangle - 2\langle v, w \rangle$.
9. Consider $u = (1, 2, 4), v = (2, -3, 5), w = (4, 2, -3)$ in $\mathbb{R}^3$, find $u \cdot v, u \cdot w, v \cdot w, (u + v) \cdot w, \|u\|$ and $\|v\|$.
10. Consider $f(t) = 3t - 5, g(t) = t^2$ in $P(t)$ then the inner product is defined as $\langle p, q \rangle = \int_0^1 p(t)q(t)dt$. Find $\langle f, g \rangle, \|f\|$ and $\|g\|$.
11. Consider $f(t) = t + 2, g(t) = 3t - 2$ and $h(t) = t^2 - 2t - 3$ in $P(t)$ with the inner product defined as $\langle p, q \rangle = \int_0^1 p(t)q(t)dt$. Find $\langle f, g \rangle, \langle f, h \rangle, \|f\|$ and $\|g\|$. Normalize f and g .
12. If $u = (1, 3, -4, 2), v = (4, -2, 2, 1)$ in $\mathbb{R}^4$ then prove the following:
 - a) triangular inequality: $\|u + v\| \leq \|u\| + \|v\|$,
 - b) Schwartz inequality: $\langle u, v \rangle^2 \leq \langle u, u \rangle \langle v, v \rangle$ or $|\langle u, v \rangle| \leq \|u\| \|v\|$.
 - c) Parallelogram law: $\|u + v\|^2 + \|u - v\|^2 = 2(\|u\|^2 + \|v\|^2)$
13. Find the angle between the vectors
 - a) $u(2, 3, 5)$ and $v = (1, -4, 3)$ in $\mathbb{R}^3$ w.r.t. the usual inner product.
 - b) $u = (1, 3, -5, 4)$ and $v = (2, -3, 4, 1)$ in $\mathbb{R}^4$.
 - c) $A = \begin{bmatrix} 9 & 8 & 7 \\ 6 & 5 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ where $\langle A, B \rangle = \text{Tr}(B^T A)$.
 - d) $f(t) = 3t - 5$ and $g(t) = t^2$ in $P(t)$ with $\langle p, q \rangle = \int_0^1 p(t)q(t)dt$.



14. Check whether $f(t) = \sin(t)$ and $g(t) = \cos(t)$ in $C[a, b]$ with $\langle f, g \rangle = \int_{-\pi}^{\pi} f(t)g(t)dt$ are orthogonal.
15. In vector space $P(x)$ with the inner product defined as $\langle p, q \rangle = \int_0^1 p(x)q(x)dx$, determine the constants a and b such that the polynomial $p(x) = x^2 + ax + b$ is orthogonal to both 1 and x .
16. Consider a matrix $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 5 & 4 \\ 1 & 4 & 6 \end{bmatrix}$ and the vectors $u = \begin{bmatrix} 1 \\ \alpha \\ 1 \end{bmatrix}$ and $v = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$. The inner product $\langle u, v \rangle = u^T Av$. Find the value of α if it is known that the vectors u and v are orthogonal and hence find the length of u .
17. Find a nonzero vector that is orthogonal to $(1,2,1)$ and $(2,5,4)$ in $\mathbb{R}^3$.

II. Orthogonal complements and orthogonal basis

1. Let W be a subspace of $\mathbb{R}^5$ spanned by $u = (1,2,3,-1,2)$ and $v = (2,4,7,2,-1)$. Find the basis of $W^\perp$.
2. Find a basis of W of $\mathbb{R}^4$ orthogonal to $u_1 = (1,-2,3,4)$ and $u_2 = (3,-5,7,8)$.
3. Show that $S = \{(1,1,0,-1), (1,2,1,3), (1,1,-9,2), (16,-13,1,3)\}$ is orthogonal and a basis of $\mathbb{R}^4$. Find the coordinates of the arbitrary vector $v = (a,b,c,d)$ in $\mathbb{R}^4$ relative the basis S .
4. Show that $S = \{(1,1,1,1), (1,1,-1,-1), (1,-1,1,-1), (1,-1,-1,1)\}$ is orthogonal and a basis of $\mathbb{R}^4$. Express $v = (1,3,-5,6)$ as a linear combination of the vectors of S . Find the coordinates of the arbitrary vector $v = (a,b,c,d)$ in $\mathbb{R}^4$ relative the basis S .
5. $P_2(t)$ is the vector space of polynomials of degree ≤ 2 with $\langle f, g \rangle = \int_0^1 f(t)g(t)dt$. Find a basis of the subspace W orthogonal to $h(t) = 2t + 1$.
6. Let $A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 3 & -1 \\ 4 & 6 & 5 \end{bmatrix}$. Find the basis of orthogonal complement of row space of A .

III. Projections

1. Find the projection of the vector v along w or $W = \text{span}(w_i)$:
 - a) $v = (1,-2,3,4)$ along $w = (1,2,1,2)$ in $\mathbb{R}^4$.
 - b) $v = (1,3,1,2)$ along $w = (1,-2,7,4)$ in $\mathbb{R}^4$.
 - c) $v = (1,2,3,4,6)$ along $W = \text{span}\{(1,2,1,2,1), (1,-1,2,-1,1)\}$ in $\mathbb{R}^5$.
 - d) $v = (1,3,5,7)$ along $W = \text{span}\{(1,1,1,1), (1,-3,4,-2)\}$ in $\mathbb{R}^4$.
 - e) $v = t^2$ along $w = t + 3$ in $P(t)$ with $\langle f, g \rangle = \int_0^1 f(t)g(t)dt$.
 - f) $v = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ along $w = \begin{bmatrix} 1 & 1 \\ 5 & 5 \end{bmatrix}$ in $M_{2 \times 2}$ with $\langle A, B \rangle = \text{tr}(B^T A)$.

Gram-Schmidt Process

Suppose $\{v_1, v_2, \dots, v_n\}$ is a basis of an inner product space V . One can use this basis to construct an orthogonal basis $\{w_1, w_2, \dots, w_n\}$ of V as follows. Set

$$w_1 = v_1$$

$$w_k = v_k - \sum_{i=1}^{k-1} c_{ki} w_i = v_k - \sum_{i=1}^{k-1} \text{proj}(v_k, w_i)$$

where $c_{ki} = \frac{\langle v_k, w_i \rangle}{\langle w_i, w_i \rangle}$ is the component of v_k along w_i .



IV. Gram-Schmidt orthogonalization

- Find an orthogonal basis and hence an orthonormal basis of the subspace W spanned by the following vectors
 - $v_1 = (1,1,1,1)$, $v_2 = (1,2,4,5)$ and $v_3 = (1, -3, -4, -2)$ of $\mathbb{R}^4$.
 - $v_1 = (1,1,1,1)$, $v_2 = (1,1,2,4)$ and $v_3 = (1,2, -4, -3)$ of $\mathbb{R}^4$.
 - $v_1 = (1,1,1,1)$, $v_2 = (1, -1,2,2)$ and $v_3 = (1,2, -3, -4)$ of $\mathbb{R}^4$.
 - $S = \{1, t, t^2, t^3\}$ of $P_3(t)$ given $\langle f, g \rangle = \int_{-1}^1 f(t)g(t)dt$.
- Find an orthogonal basis for the subspace $w^\perp$ of $\mathbb{R}^4$, where $w = (1,2,3,1)$.
- Find an orthogonal basis for the subspace $w^\perp$ of $\mathbb{R}^4$, where $w = (1, -2, -1,3)$.
- Find an orthogonal matrix P whose first row is $u_1 = (\frac{1}{3}, \frac{2}{3}, \frac{2}{3})$.
- Find an orthogonal matrix P whose first two rows are linear combination of $u_1 = (1,1,1)$ and $u_2 = (1, -2,3)$.
- Find an orthonormal basis of $P_2(\mathbb{R})$, where the inner product is given by $\langle p, q \rangle = \int_0^1 p(x)q(x)dx$.
- Let $M_{2 \times 2}$ be the vector space of real matrices with inner product defined as $\langle A, B \rangle = \text{tr}(B^T A)$. Consider the matrices $V = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and $W = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$, apply the Gram-Schmidt orthogonalization process to the matrices with respect to this inner product to obtain an orthogonal and orthonormal set of matrices.

Steps for QR Factorization (Using Gram-Schmidt Process)

For a given matrix A of size $m \times n$ (where $m \geq n$):

- Extract Column Vectors:

Let $A = [a_1, a_2, \dots, a_n]$, where a_i are the column vectors of A .

- Apply Gram-Schmidt Orthogonalization:

Construct an orthonormal set of vectors $q_1, q_2, \dots, q_n$ from $a_1, a_2, \dots, a_n$

- Compute the first orthonormal vector: $q_1 = \frac{a_1}{\|a_1\|}$
- For $i = 2$ to n , compute u_i using Gram-Schmidt orthogonalization then normalize:

$$q_i = \frac{u_i}{\|u_i\|}$$

- Construct Matrices Q and R :

- Q is formed by placing $q_1, q_2, \dots, q_n$ as its columns.
- R is computed as:

$$R = Q^T A$$

where R is an upper triangular matrix.

V. QR factorization

- Obtain the QR factorization of the following matrices:

$$\text{a) } A = \begin{bmatrix} 1 & -1 & -1 \\ 1 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \end{bmatrix}$$

$$\text{b) } A = \begin{bmatrix} 2 & -1 \\ -1 & 3 \\ 0 & 1 \end{bmatrix}$$

$$\text{c) } A = \begin{bmatrix} 1 & 0 & 2 \\ -1 & 2 & 0 \\ -1 & -2 & 2 \end{bmatrix}$$

$$\text{d) } A = \begin{bmatrix} 1 & 0 & -1 \\ 2 & -3 & 3 \\ -1 & 2 & 4 \end{bmatrix}$$

$$\text{e) } A = \begin{bmatrix} 1 & 2 \\ -1 & -2 \\ 1 & 1 \end{bmatrix}$$

- Solve the system of equations $AX = B$ where $A = \begin{bmatrix} 2 & 3 \\ 2 & 4 \\ 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 7 \\ 3 \\ 1 \end{bmatrix}$ by QR factorization method.



3. Solve the system of equations $AX = B$ where $A = \begin{bmatrix} 3 & -2 \\ 2 & -3 \\ 1 & -1 \\ 2 & 3 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 2 \\ -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$ by QR factorization method.

Working rule of Least square method

1. Multiply A^T on both sides of the given system to get $A^T A X = A^T b$
2. Simplify $X = (A^T A)^{-1} A^T b$ if $A^T A$ is invertible and of lower order to get least square solution X
3. If $A^T A$ is not invertible then solve $A^T A X = A^T b$ using Gauss elimination method to get least square solution X
4. The least square error is given by $\|AX - B\|$.

Note: Given $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ the coefficient matrix A and constant matrix b of

the system of equation $AX = b$ are given by $A = \begin{bmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ \vdots & \vdots & \vdots \\ 1 & x_n & x_n^2 \end{bmatrix}$ and $b = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$.

VI. Least square method

1. Solve the following system of equations $AX = b$ by the method of least squares and hence find the least square error:

a) $A = \begin{bmatrix} 2 & 1 \\ 1 & 0 \\ 0 & -1 \\ -1 & 1 \end{bmatrix}$ and $b = \begin{bmatrix} 3 \\ 1 \\ 2 \\ -1 \end{bmatrix}$. ii) $A = \begin{bmatrix} 3 & -2 \\ 2 & -3 \\ 1 & -1 \\ 2 & 3 \\ 3 & 4 \end{bmatrix}$ and $b = \begin{bmatrix} 2 \\ -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$.

2. In an experiment designed to determine the extent of a person's natural orientation, a subject is put in a special room and kept there for a certain length of time. He is then asked to find a way of a maze and record is made of the time it takes the subject to accomplish this task. The following data are obtained.

Time in Room (hours)	1	2	3	4	5	6
Time to find Way out of maze(minutes)	0.8	2.1	2.6	2.0	3.1	3.3

Let x denote the number of hours in the room and let y denote the number of minutes that it takes the subject to find his way out.

- a) Find the least squares line of the form $y = a + bx$.
 - b) Use the equation obtained in (a) to estimate the time it will take the subject to find his way out of the maze after 10 hours in the room.
 - c) Compute the least-squares error correct to 3 decimal places.
3. Find the least squares line of the form $y = a + bx$ for the data $(0, -1), (1, 3), (2, 5)$ and $(3, 7)$.
 4. Find the equation $y = a + bx + cx^2$ of the least-square curve that best fits the data points $(2, 1), (5, 2), (7, 3)$ and $(8, 3)$.
 5. Find the least square curve $y = a + bx + cx^2$ for the data $(0, 1), (1, 1.8), (2, 1.3), (3, 2.5)$ and $(4, 6.2)$.
 6. A steel producer gathers the following data.

Year	1997	1998	1999	2000	2001	2002
Annual Sales (millions of dollars)	1.2	2.3	3.2	3.6	3.8	5.1



Represent the years 1997, ..., 2002 as 0, 1, 2, 3, 4, 5 respectively and let x denote the year. Let y denote the annual sales (in millions of dollars).

- Find the least squares line of the form $y = ax + b$.
- Use the equation obtained in (a) to estimate the annual sales for the year 2006.

7. A sales organisation obtains the following data relating the number of salespersons (x) to annual sales (y).

Number of salespersons	5	6	7	8	9	10
Annual Sales (millions of dollars)	2.3	3.2	4.1	5.0	6.1	7.2

Let x denote number of salesperson and y denote the annual sales (in millions of dollars).

- Find the least squares curve of the form $y = a + bx + cx^2$.
- Use the equation in (a) to estimate the annual sales when there are 15 salespersons.